

Spot Rates vs. Forward Rates: Mini Study Guide

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A spot rate is the annual return from today to year n ; a forward rate is the rate for one future year that today's curve implies. This guide shows how to get each from zero-coupon prices, why they differ, and what today's US curve says.

Learning objectives

- Define a spot rate and a forward rate, and say which period each one covers.
- Compute spot rates and forward rates from zero-coupon prices.
- Explain how to lock in a forward rate with a synthetic forward loan.
- Predict whether forwards sit above or below spots from the slope of the curve.
- Price a bond with spot rates or forward rates and get the same answer.

Core concept: a spot rate is an average, a forward rate is one leg

A spot rate y_n is the average annual return from today to year n , like the average speed for a whole trip. A forward rate f_n is the rate for the single year from $n-1$ to n , like the speed on one leg of that trip.

The spot rate is the geometric average of the forward rates up to year n :

$$(1 + y_n)^n = (1 + f_1)(1 + f_2) \cdots (1 + f_n)$$

Solving for one forward rate gives the formula to use when you are handed spot rates:

$$f_n = \frac{(1 + y_n)^n}{(1 + y_{n-1})^{n-1}} - 1$$

With zero-coupon bonds of the same face value, a shortcut avoids the yields entirely. Divide the price of the shorter bond by the price of the longer one:

$$1 + f_n = \frac{P_{n-1}}{P_n}$$

For a forward covering several years, from year a to year b :

$$f_{a,b} = \left[\frac{(1 + y_b)^b}{(1 + y_a)^a} \right]^{1/(b-a)} - 1$$

	Spot rate y_n	Forward rate f_n
Period covered	Today to year n	One year, from $n-1$ to n
Where it comes from	The yield on an n -year zero-coupon bond	Two spot rates, or two zero-coupon prices
Year 1	y_1	$f_1 = y_1$, the only year with no earlier leg
Pricing use	Discount factor $1 \div (1 + y_n)^n$	Discount factor $1 \div [(1 + f_1)(1 + f_2) \cdots (1 + f_n)]$

The slope of the spot curve fixes the slope of the forwards. If spot rates rise with maturity, forward rates sit above them; if the curve is inverted, forwards sit below; if it is flat, they are equal.

Worked example 1: spot and forward rates from zero-coupon prices

Zero-coupon bonds with \$1,000 face value trade at the prices below. The spot rate for each maturity comes from its price, and each forward rate from the ratio of adjacent prices.

Maturity (years)	Price	Spot rate y_n	Forward rate f_n
1	\$925.93	8.00%	8.00%
2	\$853.39	8.25%	8.50%
3	\$782.92	8.50%	9.00%
4	\$715.00	8.75%	9.50%
5	\$650.00	9.00%	10.00%

Spot rate. Solve the price of a zero for its yield. For the 3-year bond:

$$y_3 = \left(\frac{1000}{782.92} \right)^{1/3} - 1 = 8.50\%$$

Forward rate, two ways. The price ratio gives the year-4 forward rate directly, and the spot-rate formula agrees:

$$1 + f_4 = \frac{782.92}{715.00} = 1.095 \Rightarrow f_4 = 9.50\%$$

$$f_4 = \frac{(1.0875)^4}{(1.085)^3} - 1 = 9.50\%$$

The forward rate rises every year because the spot curve slopes upward, and each forward is above the spot rate for its own maturity.

Worked example 2: lock in a forward rate with a synthetic loan

You can lock in a forward rate today by issuing one zero and buying a longer one. To lend for one year beginning at the end of year 3, which is the year-4 forward rate, issue a 3-year zero and buy 4-year zeros with the proceeds.

The number of 4-year zeros per 3-year zero issued is the price ratio:

$$\frac{782.92}{715.00} = 1.095 \text{ four-year zeros}$$

Time	Cash flow	What happens
0	\$0	Proceeds from the 3-year zero pay for the 4-year zeros
3	-\$1,000	The 3-year zero matures; you pay its face value
4	+\$1,095	The 4-year zeros mature; you receive their face value

You receive nothing at time 0, pay \$1,000 at time 3, and receive \$1,095 at time 4. The 9.5% return on that one-year loan is exactly f_4 .

For a forward loan beginning at the end of year 4, which is the year-5 rate, repeat with the 4-year and 5-year zeros:

$$\frac{715.00}{650.00} = 1.100 \text{ five-year zeros} \Rightarrow f_5 = 10.0\%$$

The cash flows are $-\$1,000$ at time 4 and $+\$1,100$ at time 5.

Today's US curve: par, spot, and forward rates

On Oct 5, 2026, quoted Treasury strip yields implied a 1-year spot rate of 4.43%, a 10-year spot rate of 5.48%, and a one-year forward rate for year 10 of about 5.76%.

The spot rates are market quotes: ask yields on Treasury STRIPS, which are zero-coupon bonds, from a Schwab account export dated Oct 5, 2026. The par yields are Treasury's official curve for Oct 2, the latest published, so the two columns are three days apart.

Year	Par yield (Treasury, Oct 2)	Spot rate (Treasury strips, Oct 5)	One-year forward rate for that year
1	4.46%	4.43%	4.43%
2	4.83%	4.79%	5.16%
3	4.96%	5.00%	5.43%
4	5.01%*	5.12%	5.45%
5	5.06%	5.19%	5.47%
6	5.12%*	5.26%	5.63%
7	5.17%	5.32%	5.69%
8	5.21%*	5.39%	5.84%
9	5.24%*	5.45%	5.94%
10	5.28%	5.48%	5.76%

*Interpolated between the quoted 3-, 5-, 7-, and 10-year par yields. Strip yields are ask yields on interest (coupon) strips, converted to annual compounding and interpolated to whole-year maturities. Forwards are computed from unrounded spot rates.

Four readings of the table:

- **Forwards sit above spots.** The curve slopes upward, so the implied 1-year rate for year 2 is 5.16%, 73 basis points above the 1-year spot rate of 4.43%.
- **The gap is expectations plus a premium.** Under the expectations hypothesis the whole rise is expected rate hikes; under liquidity preference part of it is a term premium.
- **A multi-year forward.** The implied average rate for years 6 to 10, the 5y5y forward, is about 5.77%.
- **Spot rates run above par yields at long maturities.** The 10-year spot rate is 5.48% against a 5.28% par yield. Part of that gap is the upward slope, which pushes spot rates above par yields, and part is the three-day difference in dates.

Principal strips quoted lower yields than the interest strips used here, about a quarter of a percentage point lower at 10 years. The table uses interest strips because every payment date has a quote, while many principal strips had none.

Schwab's ask yields include the dealer's markup and move during the day; these were pulled on Oct 5, 2026. The list ends at Nov 2036, so the 10-year point rests on the last few strips.

Cross-check: quoted STRIPS yields at the long end

Beyond the 10-year strips in the Schwab list, the only free, dated STRIPS yield found covers 25+ year maturities. The [iShares 25+ Year Treasury STRIPS ETF](#) held principal STRIPS with an average yield to maturity of 5.71% on Oct 2, 2026, at an average maturity of 27.4 years.

A 27-year spot rate bootstrapped from the Oct 2 par curve is about 5.77% compounded annually, or about 5.68% on the semiannual basis that STRIPS yields use. That is within 3 basis points of the fund's 5.71%. Shorter strips are the Schwab quotes above.

Sources: Treasury par yields for Oct 2, 2026, as published at [Slickcharts](#) from Treasury.gov [Daily Treasury Par Yield Curve Rates](#). Treasury strip ask prices from a Schwab online account export of US Treasury Zeros dated Oct 5, 2026; spot and forward rates are calculated from those prices.

BA II Plus keystrokes

Set P/Y to 1 first: [2nd] [P/Y], type 1, [ENTER], [2nd] [QUIT]. Keys are in brackets, and a second function is written [2nd] [KEY].

Task	Keys	Check (Example 1)
Forward rate from two zero prices	$P(n-1) [\div] P(n) [-] 1 [=]$	782.92 $[\div]$ 715 $[-]$ 1 $[=]$ gives 0.0950
Spot rate from a zero price	[2nd] [CLR TVM], n [N], price $[+/-]$ [PV], 1000 [FV], [CPT] [I/Y]	n = 3, price 782.92 gives 8.50
Forward from two spot rates	1.085 $[y^x]$ 3 $[=]$ [STO] 1, then 1.0875 $[y^x]$ 4 $[\div]$ [RCL] 1 $[-]$ 1 $[=]$	gives 0.0950
Several-year forward from prices	$P(a) [\div] P(b) [=]$ $[y^x]$ (b-a) $[1/x]$ $[=]$ $[-]$ 1 $[=]$	5y5y: use the 5- and 10-year prices

The calculator evaluates keys from left to right, so in the third row the denominator is stored first and the division comes last. In the fourth row, type b-a as one number (5 for the 5y5y forward), then press [1/x].

Common pitfalls

- **Calling f_3 the 3-year rate.** f_3 covers year 3 only. The 3-year rate is the spot rate y_3 .
- **Discounting a cash flow at a single forward rate.** Chain every forward from year 1: the year-3 factor is $1 \div [(1+f_1)(1+f_2)(1+f_3)]$.
- **Using the par yield as a spot rate.** Par yields blend several cash flows; bootstrap them first when precision matters.
- **Expecting forwards to be forecasts.** A forward is the rate you can lock in today, and it can include a term premium.
- **Mixing up the price ratio.** Use the shorter bond's price on top: $P(n-1) \div P(n)$, not the reverse.
- **Forgetting that $f_1 = y_1$.** The first year has no earlier leg, so spot and forward are the same.

Key terms checklist

Check each term off when you can define it in your own words.

- ☐ **Spot rate:** the annual yield on a zero-coupon bond from today to a given maturity.
- ☐ **Forward rate:** the rate for one future year implied by today's spot curve.
- ☐ **Zero-coupon bond:** a bond that pays only its face value at maturity.

- ☐ **Par yield:** the coupon rate at which a bond prices at its face value.
- ☐ **Bootstrapping:** extracting spot rates from par yields one maturity at a time.
- ☐ **Geometric average:** the average rate that compounds to the same total growth.
- ☐ **Synthetic forward loan:** a loan locked in today by issuing one zero and buying a longer one.
- ☐ **Expectations hypothesis:** forward rates equal the expected future short rates.
- ☐ **Liquidity (term) premium:** extra yield for lending long, which makes forwards exceed expected rates.
- ☐ **Upward-sloping curve:** spot rates rise with maturity, so forwards lie above spots.

CFA-style questions

1. The 1-year spot rate is 5% and the 2-year spot rate is 6%. The one-year forward rate for year 2 is closest to:
 - A. 5.5%
 - B. 6.0%
 - C. 7.0%
2. Which statement about forward rates is correct?
 - A. The year-3 forward rate is the annual return from today to year 3.
 - B. Under the expectations hypothesis, forward rates equal expected future short rates.
 - C. The year-1 forward rate is always higher than the 1-year spot rate.
3. Zero-coupon bonds with \$1,000 face value trade at \$782.92 for 3 years and \$715.00 for 4 years. The forward rate for year 4 is closest to:
 - A. 8.75%
 - B. 9.50%
 - C. 10.00%
4. If the spot curve slopes upward, the forward curve lies:
 - A. above the spot curve.
 - B. below the spot curve.
 - C. on top of the spot curve.
5. A 3-year bond is priced once with the spot rates and once with the matching forward rates. The two prices are:
 - A. higher with the forward rates.
 - B. the same.
 - C. lower with the forward rates.
6. On Oct 5, 2026, the 1-year spot rate was 4.43% and the implied forward rate for year 2 was 5.16%. This most likely means the market expects, or demands compensation for:
 - A. falling short-term rates.
 - B. rising short-term rates.
 - C. an unchanged short-term rate.

Answer key

Q	Answer	Explanation
1	C	$f_2 = (1.06)^2 \div 1.05 - 1 = 1.1236 \div 1.05 - 1 = 7.01\%$. Averaging 5% and 6% (A or B) ignores compounding.

Q	Answer	Explanation
2	B	The expectations hypothesis sets forwards equal to expected future short rates. A describes a spot rate, and f_1 always equals y_1 , so C is false.
3	B	$1 + f_4 = 782.92 \div 715.00 = 1.095$, so $f_4 = 9.50\%$. A is the 4-year spot rate, and C is the year-5 forward rate.
4	A	Each forward is the marginal rate for one year. When the average (spot) rate is rising, the marginal rate must exceed it.
5	B	The spot rate is the geometric average of the forwards, so each discount factor is identical and so is the price.
6	B	A forward above the spot rate says the 1-year rate is expected to rise, or that lenders demand a term premium. Both reflect a rising path.